



BAULKHAM HILLS HIGH SCHOOL

Half -Yearly 2014
YEAR 11 ADVANCED TASK 1

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 1.5 hours
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working in Questions 11-15
- Marks may be deducted for careless or badly arranged work

Total marks – 64

Exam consists of 6 pages.

This paper consists of TWO sections.

Section 1 – (10 marks) Pages(2-3)

Questions 1-10

- Attempt Question 1-10
- Answer on answer sheet provided

Section II – (54 marks) Pages(4-6)

- Attempt questions 11-15

Section I - 10 marks**Use the multiple choice answer sheet for question 1-10**

1. 0.01208 in scientific notation to 3 significant figures is:
- (A) 0.0121
(B) 0.121×10^{-1}
(C) 1.20×10^{-2}
(D) 1.21×10^{-2}
2. Council rates increase by 4% to \$1400. What were rates before the increase, to the nearest dollar?
- (A) \$56
(B) \$1344
(C) \$1346
(D) \$1456
3. The solutions to the equation $x^2 - 7x - 2 = 0$ are :
- (A) $\frac{7 \pm \sqrt{41}}{2}$
(B) $\frac{7 \pm \sqrt{57}}{2}$
(C) $\frac{-7 \pm \sqrt{41}}{2}$
(D) $\frac{-7 \pm \sqrt{57}}{2}$
4. Which of the following is equivalent to $\frac{1}{2\sqrt{5} - \sqrt{3}}$?
- (A) $\frac{2\sqrt{5} - \sqrt{3}}{7}$
(B) $\frac{2\sqrt{5} + \sqrt{3}}{7}$
(C) $\frac{2\sqrt{5} - \sqrt{3}}{17}$
(D) $\frac{2\sqrt{5} + \sqrt{3}}{17}$
5. The solution to the equation $1 + \frac{1-3x}{2} \geq 2x$ is :
- (A) $x \leq \frac{2}{7}$
(B) $x \geq \frac{2}{7}$
(C) $x \leq \frac{3}{7}$
(D) $x \geq \frac{3}{7}$

6. If $\sin \theta = \frac{3}{7}$ and $\cos \theta < 0$ then $\cot \theta$ is

- (A) $\frac{3}{\sqrt{40}}$
- (B) $-\frac{3}{\sqrt{40}}$
- (C) $\frac{\sqrt{40}}{3}$
- (D) $-\frac{\sqrt{40}}{3}$

7. The solution to the equation $|2x + 2| = x - 3$ is:

- (A) $x = -5$ and $x = \frac{1}{3}$
- (B) $x = -5$
- (C) $x = \frac{1}{3}$
- (D) no solution

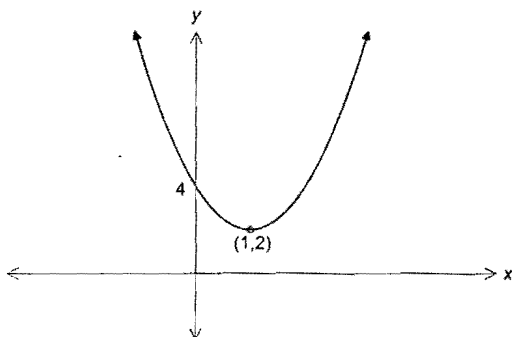
8. $\frac{8^{n+1}}{2^{n-2}} =$

- (A) 4^{-1}
- (B) 4^3
- (C) 2^{2n+1}
- (D) 2^{2n+5}

9. $f(x) = \cos x$ is

- (A) an even function
- (B) an odd function
- (C) neither odd nor even
- (D) not a function

10.



The equation for the given parabola is:

- (A) $y = (x + 1)^2 + 2$
- (B) $y = (x - 1)^2 + 2$
- (C) $y = (x - 2)^2 + 1$
- (D) $y = 2(x - 1)^2 + 2$

End of Section 1

Section II – Extended Response

Attempt questions 11-15. All necessary working should be shown in every question.

Question 11 (12 marks)

Marks

- a) Express $0.2\dot{3}$ as a fraction in its simplest form. 1
- b) Expand and simplify $(3\sqrt{2} - \sqrt{7})^2 =$ 2
- c) If $7\sqrt{2} - \frac{\sqrt{8}}{2} = \sqrt{a}$ find a . 2
- d) Factorise (i) $x^2 - 5x - 6$ 1
(ii) $2x^3 - 16$ 2
(iii) $x^3 - x^2 - 9x + 9$ 2
- e) Find the exact value of $\sin(240^\circ)$ 2

End of Question 11

Question 12 (11 marks)

- a) Simplify $\frac{1}{x^2 - 4} + \frac{1}{2x + 4}$ 2
- b) Solve (i) $|3x - 1| < 5$ 2
- (ii) $\frac{x+4}{2} - \frac{x+3}{4} = 10$ 2
- c) Simplify $\frac{ab^{-1} - ba^{-1}}{b - a}$ 2
- d) 420 adults and children attended a school musical. There were 120 more children than adults. Form a pair of simultaneous equations and solve them to determine the number of adults and children that attended the musical. 3

End of Question 12

Question 13 (11 marks)**Marks**

a) Sketch each of the following showing all essential features.

(i) $y = 4 - |x|$

2

(ii) $x^2 + (y - 2)^2 = 4$

2

(iii) State the range for the graphs in part (i) and (ii).

2

b) State the domain for the function $y = \sqrt{x^2 - 9}$.

2

c) Sketch the graph of $y = 2 - \frac{1}{x-1}$

3

End of Question 13**Question 14 (10marks)**

a) Shade the region which satisfies both of the following inequalities :

$$y \leq \sqrt{4 - x^2} \text{ and } y \geq x - 2.$$

3

b) Solve the following if $0^\circ < \theta < 360^\circ$

(i) $2 \sin \theta - 1 = 0$

2

(ii) $\tan^2 \theta = 2 \sec \theta + 2$

3

c) If $f(x+1) = \frac{2f(x)+3}{3}$ and $f(3) = 4$, find $f(2)$.

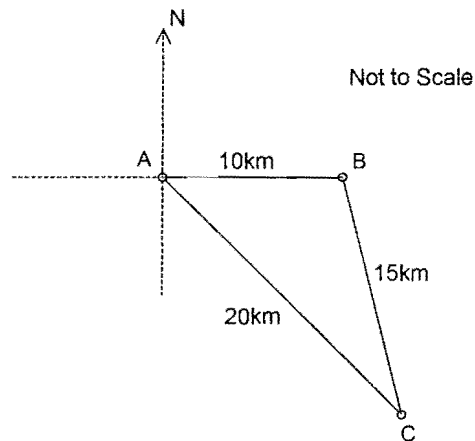
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End of Question 14

Question 15 (10 marks)

Marks

- a) Blake's journey is shown in the diagram below. He walks 10 km due east from A to B, 15 km from B to C then 20 km back to A.

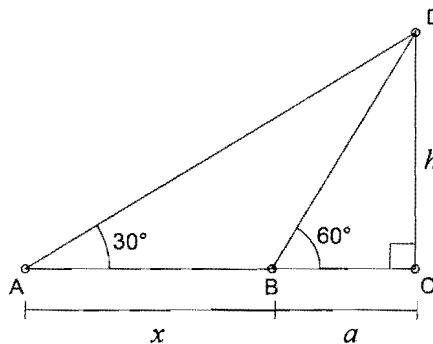


- (i) Find $\angle ABC$
- (ii) Hence state the bearing of B from C.

2

1

b)



Given the above diagram:

- (i) Show that $h = \frac{x+a}{\sqrt{3}}$
- (ii) Hence or otherwise find an expression for x in terms of a which is independent of h .

1

2

- c) If k is a solution to $x^2 - x - 1 = 0$. Show that

(i) $k^2 = k + 1$

(ii) $k^6 = 8k + 5$

1

3

End of Exam

Solutions. (Total /64)

1. D 2. C 3. B 4. D 5. C
6. D 7. D 8. D 9. A 10. D

11. a) let $x = 0.2333$

10x = 2.333
100x = 23.333

$$\therefore 90x = 21 \text{ (accept } \frac{21}{90})$$

$$x = \frac{1}{30} \quad (1)$$

b) $(3\sqrt{2} - \sqrt{7})^2$
 $= 18 - 6\sqrt{14} + 7$ 1 off each error
 $= 25 - 6\sqrt{14} \quad (2)$

c) $7\sqrt{2} - \frac{\sqrt{8}}{2} = 7\sqrt{2} - \frac{2\sqrt{2}}{2}$
 $= 6\sqrt{2}$
 $= \sqrt{72}$

$$\therefore a = 72. \quad (1)$$

d) (i) $x^2 - 5x - 6$
 $= (x-6)(x+1) \quad (1)$

(ii) $2x^3 - 16 = 2(x^3 - 8)$
 $= 2(x-2)(x^2 + 2x + 4) \quad (1)$

(ii) $x^3 - 9x - x^2 + 9$
 $= x(x^2 - 9) - 1(x^2 - 9)$
 $= (x-1)(x^2 - 9)$
 $= (x-1)(x+3)(x-3) \quad (1)$

e) $\sin 240 = -\sin 60$
 $= -\frac{\sqrt{3}}{2} \rightarrow (1)$

12. (ii) $\frac{1}{x^2-4} + \frac{1}{2x+4}$
 $= \frac{1}{(x+2)(x-2)} + \frac{1}{2(x+2)}$
 $= \frac{2 + x-2}{2(x+2)(x-2)}$
 $= \frac{x}{2(x+2)(x-2)} \quad (1)$

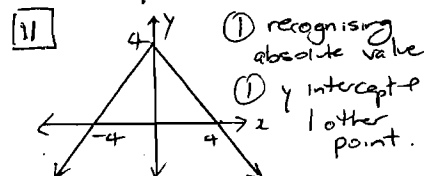
b) (i) $|3x-1| < 5$ or $3x-1 < 5$
 $-5 < 3x-1 < 5$
 $-4 < 3x < 6$
 $-\frac{4}{3} < x < 2$
 $x < 2$
 $-3x+1 < 5$
 $-3x < 4$
 $x > -\frac{4}{3}$
 $\therefore -\frac{4}{3} < x < 2$
 (must write as 1 expression)

13. (i) $\frac{x+4}{2} - \frac{x+3}{4} = 10$
 $2x+8 - x-3 = 40$
 $x+5 = 40$
 $x = 35 \quad (1)$

c) $\frac{ab^{-1} - ba^{-1}}{b-a}$ If $\frac{1}{ab} - \frac{1}{ba}$
 $\frac{\frac{a}{b} - \frac{b}{a}}{b-a}$ the answer is 0 - a word
 (1) only
 $= \frac{a^2 - b^2}{ab(b-a)}$
 $= \frac{(a+b)(a-b)}{ab(b-a)} \times \frac{1}{(b-a)}$
 $= \frac{-a-b}{ab} \quad (1)$

d) $a+c = 420 \quad (1)$
 $c = a+120 \quad (2)$
 sub (2) into (1)
 $a+a+120 = 420$
 $2a = 300$
 $a = 150 \quad (1)$
 $c = 270 \quad (1)$
 adults - 150
 children - 270

13a) (i) $y = 4 - |x|$



(ii) $x^2 + (y-2)^2 = 4$

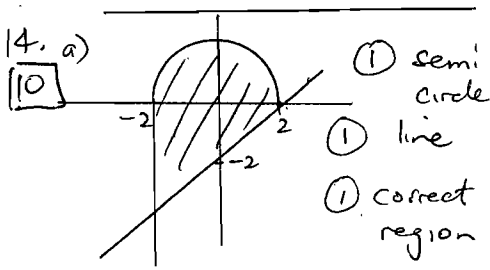
 (1) Centre
 (1) radius

(iii) Range: $y \leq 4 \quad (1)$

(iv) $R: 0 \leq y \leq 4 \quad (1)$

b) $y = \sqrt{x^2 - 9}$
 (1) $x^2 - 9 \geq 0$
 $x \leq -3$ or $x \geq 3 \quad (1)$

c)
 (1) Correct asymptotes
 (1) recognising hyperbola
 (1) Correct position & 1 other point.



b) (i) $2\sin\theta - 1 = 0$
 $\sin\theta = \frac{1}{2} \rightarrow \theta = 30^\circ, 150^\circ$

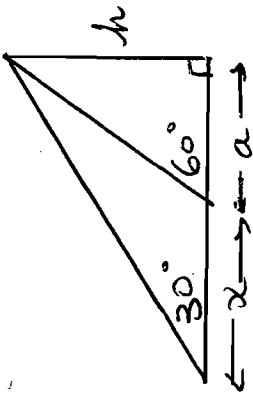
(ii) $\tan^2\theta = 2\sec\theta + 2$
 $\sec^2\theta - 1 = 2\sec\theta + 2$
 $\sec^2\theta - 2\sec\theta - 3 = 0$
 $(\sec\theta - 3)(\sec\theta + 1) = 0$
 $\sec\theta = 3, -1$
 $\therefore \cos\theta = \frac{1}{3}$
 $\cos\theta = -1$
 $\theta = 70^\circ 32', 289^\circ 28' \quad (1)$
 $\theta = 180^\circ \quad (1)$

c) $f(x+1) = \frac{2f(x)+3}{3}$
 $f(3) = 4$
 $f(3) = f(2+1) = \frac{2f(2)+3}{3}$
 $\therefore 4 = \frac{2f(2)+3}{3} \quad (1)$
 $2f(2) = 9$
 $f(2) = 4.5 \quad (1)$

15a)
 (i) $\cos B = \frac{10^2 + 15^2 - 20^2}{2 \times 10 \times 15}$
 $= 104^\circ 29' \quad (1)$

(ii) Bearing = $360 - \theta$
 $\theta = 14^\circ 29'$
 Bearing = $360 - 14^\circ 29'$
 $= 345^\circ 31' \quad (1)$
 (Accept $345^\circ - 346^\circ$)

b)



$$(i) \tan 30^\circ = \frac{h}{x+a}$$

$$\therefore h = (x+a) \times \tan 30^\circ$$

$$= (x+a) \times \frac{1}{\sqrt{3}} \quad (1)$$

$$h = \frac{x+a}{\sqrt{3}} \quad (A)$$

$$(ii) \tan 60^\circ = \frac{h}{a}$$

$$\therefore h = a \tan 60^\circ$$

$$= a\sqrt{3} \quad (1) - (B)$$

~~(A)~~ = ~~(B)~~

$$\therefore \frac{x+a}{\sqrt{3}} = a\sqrt{3}$$

$$x+a = 3a \quad (1)$$

$$\therefore x = 2a.$$

$$c) (i) x^2 - x - 1 = 0$$

x is a root $\therefore$

$$x^2 - x - 1 = 0$$

$$\therefore x^2 = x + 1. \quad \dots (1)$$

$$(ii) x^6 = (x^2)^3$$

$$= (x+1)^3$$

$$= x^3 + 3x^2 + 3x + 1$$

$$(1) = x(x^2) + 3(x+1) + 3x + 1$$

$$= x(x+1) + 3x + 3 + 3x + 1$$

$$= x^2 + x + 6x + 4$$

$$= x + 1 + 7x + 4$$

$$(1) = 8x + 5.$$